Describe a wave

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Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Identify amplitude, wavelength, frequency and period.

SUCCESS CRITERIA - I can...

1. Measure amplitude from equilibrium.
2. Measure wavelength between matching points.
3. Connect frequency and period.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “amplitude” and “wavelength” to explain your model or result. Sentence starters are optional.

amplitude: maximum displacement from equilibrium for the wave

wavelength: distance between corresponding points on repeated waves

frequency: number of cycles each second

01 Warm-up: bring back an earlier idea

A sample warms by 3°C after a certain energy input. If absorbed energy becomes 5 times as large with the same mass and material and no phase change, find the new rise.

Word help: absorbed = taken into the specified object or system

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

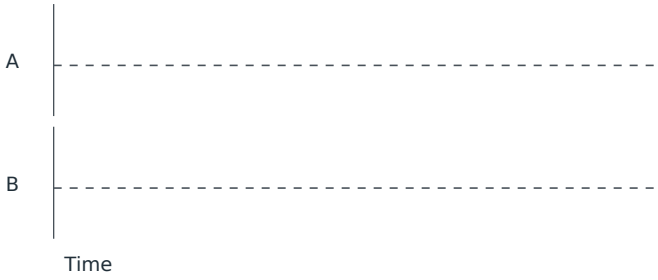
02 Worked example: explain the model

Reference: f = cycles/Δt; T = 1/f.
f is frequency (Hz); T is period (s). λ is crest-to-crest wavelength; amplitude is maximum displacement.

Worked reasoning: A wave makes 6 cycles in 3 s: f=2 Hz and period T=0.5 s. Amplitude measures maximum displacement, not the spacing of crests.

Your turn: Draw one spatial wave and label amplitude and wavelength 0. Separately mark one period on a displacement-time graph.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

Describe a wave

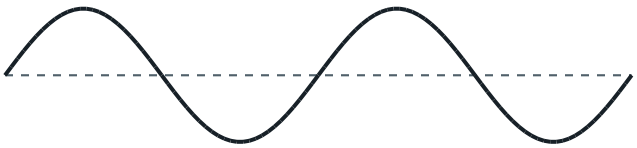
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Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Label amplitude and one wavelength on the wave diagram.
Explain why crest-to-trough height is not the amplitude.

Sentence starter: My claim is … . The diagram/data show …, so … .



Sketch and label your response.

Marzano check - circle: 1 2 3

04 Apply the physics

A point on a rope completes 6 cycles in 5 s.
Find its frequency and period. In a separate trial, each cycle takes 0.5 s.
Find that trial's frequency.
Use two time traces to show which trial oscillates more often.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: 1 2 3

Describe a wave

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Does increasing amplitude necessarily increase frequency?
Use the definitions to explain 0.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A 4 kg sample has c=600 J/(kg·°C) and warms by 8°C without changing phase.
Find Q.
Label temperatures, samples and relevant energy transfers.

Sentence starter: I found … [unit]. This value means … in this situation.

Sample A

Sample B

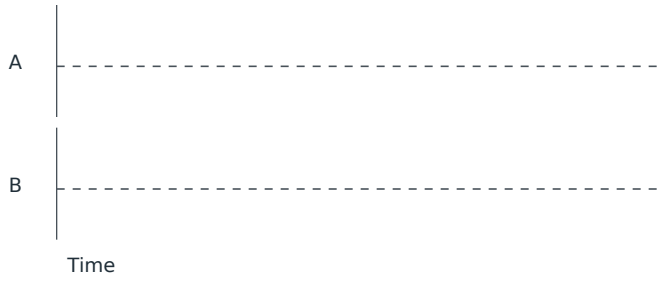
Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3

07 Exit check

A wave has period 2 s. How many cycles occur in 6 s?
Find frequency.

Sentence starter: I found … [unit]. This value means … in this situation.

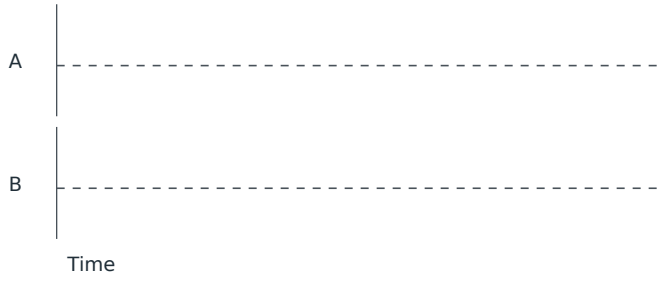


Marzano check - circle: 1 2 3

08 STRETCH: test the model

Draw two waves with equal wavelength but different amplitudes.
Explain which feature changes and which does not.

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: 1 2 3